

Rajiv Gandhi University of Health Sciences, Karnataka

Fourth Semester B. Pharm Degree Examination – 16-Jun-2022

Time: Three Hours

Max. Marks: 75 Marks

MEDICINAL CHEMISTRY - I

Q.P. CODE: 5014

Your answers should be specific to the questions asked.

Draw neat labeled diagrams wherever necessary.

All the Questions are compulsory.

LONG ESSAYS

2 x 10 = 20 Marks

1. Define and classify Anticonvulsants giving examples with one structure from each class. Outline the synthesis and uses of Carbamazepine and Phenytoin.

OR

What are Anti-inflammatory agents? Classify them with examples. Discuss their mechanism of action. Write the synthesis of Mefenamic acid.

2. What are Sympathomimetic agents? Discuss their classification and SAR. Give the synthesis of Salbutamol.

SHORT ESSAYS

7 x 5 = 35 Marks

3. Define and classify general anesthetic agents with example. Write their mode of action.

OR

Discuss the SAR of benzodiazepines.

4. Write a note on cholinesterase inhibitors. Giving examples and mode of action.

OR

What are cholinergic agents? Classify them with examples and structures.

5. Classify synthetic anticholinergics with examples. outline the synthesis and uses of Procyclidine HCl.
6. Classify alpha adrenergic blockers with examples. Write the synthesis and uses of Tolazoline.
7. Explain the role of solubility and partition co-efficient on drug action.
8. Define ligand and chelate. Explain the role of chelating agents on drugs action.
9. What is Phase-I biotransformation? Explain oxidation reactions with examples.

SHORT ANSWERS

10 x 2 = 20 Marks

10. Write a note on optical isomerism and its importance on drug action.
11. Give two examples for methylation reaction on drug metabolism.
12. Write a note on adrenergic alpha receptors.
13. Write the biosynthesis of norepinephrine.
14. What are selective and non selective beta blockers? Give one example for each.
15. Write the structure and uses of Diazepam and Glutethimide.
16. Write a note on Muscarinic receptors.
17. Write a structure and uses of Pilocarpine. Write its possible side effects.
18. Write the name and structure of two irreversible cholinesterase inhibitors.
19. Write the structure and uses of chlorpromazine and Triclofos sodium.
